SOLAR WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM

A Project Report Submitted in Partial fulfilment of the Requirements for the Award of the Degree of

BACHELOR OF ENGINEERING

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

By

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METHODIST COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution, affiliated to Osmania University & Approved by AICTE, Accredited by NAAC with A+ and NBA)

King Koti, Abids, Hyderabad, Telangana – 500001

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CERTIFICATE

This is to certify that the project titled– "**SOLAR WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM**" submitted by **G. SREEDHAR (160721734304), M. BHANU PRAKASH (1607121734318), KRISHNA CHAITHANYA (160721734005)**, is a record of students own work carried by them under my guidance and supervision in partial fulfilment for the degree of **Bachelor of Engineering** in **Electrical And Electronics Engineering** from **Methodist college of Engineering And Technology, King Koti, Abids, Hyderabad, Telangana – 500001.**

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DECLARATION

This is to declare that the work submitted in the present project work report titled **"SOLAR WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM "** is a record of Bonafide work done by me in the Department of Electrical and Electronics Engineering in Methodist College of Engineering and Technology, Abids, Hyderabad.

No part of the report is copied from books, journals, internet and wherever the subject content is taken, the same has been duly referred in the text. The report generated is based on the project work carried out entirely by us and not copied from any other source.

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Place: Hyderabad

Date:

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We are highly indebted to our parents for their constant encouragement and support during our studies.

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ABSTRACT

The objective of a solar wireless electric vehicle (EV) charging system is to provide an efficient, sustainable, and convenient method for charging electric vehicles without the need for physical connections or cables. This system aims to integrate solar power with wireless charging technology to create an eco-friendly and autonomous solution. The methodology for a solar wireless electric vehicle (EV) charging system involves several key stages, including system design, integration of components, optimization and testing. The development of a solar wireless EV charging system involves the integration of renewable energy technology with cutting-edge wireless power transfer methods. By optimizing energy generation, storage, and transfer, this methodology aims to deliver an ecofriendly, efficient, and convenient solution for the future of electric vehicle charging.
The findings from the development and evaluation of a solar wireless electric vehicle (EV) charging system can be categorized into key areas, such as performance, efficiency, challenges, and potential benefits. These findings typically come from research, prototype testing, and practical implementation. The novelty of a solar wireless electric vehicle (EV) charging system lies in the integration of two transformative technologies—solar energy and wireless power transfer (WPT)—to provide a sustainable, convenient, and efficient solution for EV charging.

Keywords: Renewable Energy, Wireless Technology, Sustainable, Integration, Prototype, Convenient, Optimization.
SDGs: SDG4 (4.7.1), SDG7 (7.1.1), SDG9 (9.4.1), SDG12 (12.2.1), SDG13 (13.2.2)

CONTENTS

LIST OF FIGURES

LIST OF TABLES

S.No. Table Name Page. No.

ABBREVIATIONS

The next list describes several symbols & abbreviation that will be later used within the body of the document

LCD	Liquid Crystal Display
RPS	Regulated Power Supply
PIC	Peripheral Interface Controller
LED	Light -Emitted Diode
PS	Power system
SP	Set point

CHAPTER 1
INTRODUCTION

1.1 Introduction:

The project aims at developing an intelligent Electric vehicle which is powered from wireless power transmission.

In transmission section we are using solar to charge the battery which is used to provide the power supply of the transmitting coil. In receiver section consist of vehicle. This vehicle receives the power from transmitting coil while moving the vehicle. Micro controller measured voltage from receiving coil will be display on LCD.

Wireless energy transfer or wireless power is the transmission of electrical energy from a power source to an electrical load without a conductive physical connection. Wireless transmission is useful in cases where interconnecting wires are inconvenient, hazardous, or impossible. The problem of wireless power transmission differs from that of wireless telecommunications, such as radio. In the latter, the proportion of energy received becomes critical only if it is too low for the signal to be distinguished from the background noise. With wireless power, efficiency is the more significant parameter. A large part of the energy sent out by the generating plant must arrive at the receiver or receivers to make the system economical. The most common form of wireless power transmission is carried out using direct induction followed by resonant magnetic induction.

Features:

- Design a power generation system.
- Using wireless power transmission technology.
- Usage of solar energy for wireless power transmission.

1.2 Project Overview:

The project “Wireless Power Transmission for Electric Vehicle Using Solar” using PIC microcontroller is an exclusive project that can transfers the power wirelessly using wireless power transmission by using solar energy and that can display the receiving voltage on LCD display.

1.3 Thesis Overview:

These explains the implementation of "**SOLAR WIRELESS ELECTRIC VEHICLE CHAGING SYSTEM"**. The organization of the thesis is explained here with:

Chapter 1 Presents introduction to the overall thesis and the overview of the project. In the project overview a brief introduction "**SOLAR WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM"** and its applications are discussed.

Chapter 2 Presents the hardware description. It deals with the block diagram of the project and explains the purpose of each block. In the same chapter the explanation of each module is considered.

Chapter 3 Presents the software description. It explains the implementation of the project using PIC C Compiler.

Chapter 4 Presents the project description along with each module interfacing.

Chapter 5 Presents the advantages, disadvantages and applications of the project.

Chapter 6 Presents the results, conclusion and future scope of the project.

Chapter 7 References

CHAPTER 2
HARDWARE DESCRIPTION

2.1 Introduction:

In this chapter the block diagram of the project and design aspect of independent modules are considered as shown in the fig: 2.1.

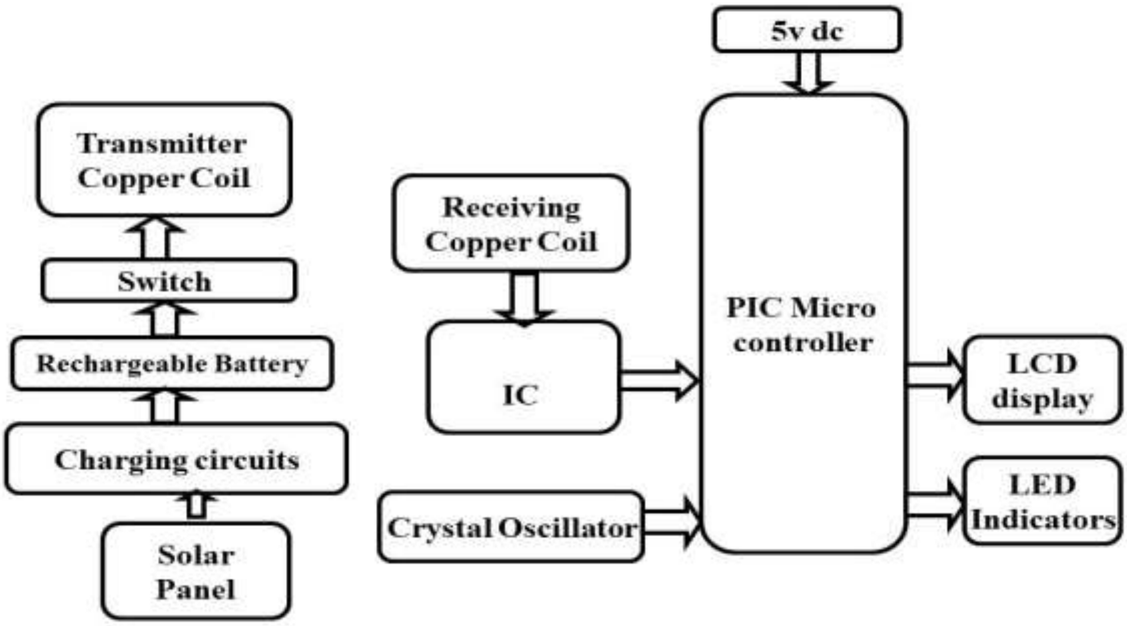

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FIG 2.1 Block diagram of Solar Wireless Electric Vehicle Charging System

The main blocks of this project are:

- Solar Panel.
- Charging circuit.
- Rechargeable battery.
- Switch.
- Transmitter Receiver charging coils.
- PIC Microcontroller.
- LCD display.
- Crystal oscillator.
- Reset button.
- LED indicator.

2.2 PIC Microcontroller:

The PIC16F72 CMOS FLASH-based 8-bit microcontroller is upward compatible with PIC16C72/72A and PIC16F872devices. It features 200 ns instruction execution, self programming, an ICD, 2 Comparators, 5 channels of 8-bit Analog-to-Digital (A/D) converter, 2 capture/compare/PWM functions, a synchronous serial port that can be configured as either 3-wire SPI or 2-wire I2C bus, a USART, and a Parallel Slave Port.

- High performance RISC CPU
- Only 35 single word instructions to learn
- All single cycle instructions except for program branches which are two-cycle
- Operating speed: DC - 20 MHz clock input DC - 200 ns instruction cycle
- 2K x 14 words of Program Memory

128 x 8 bytes of Data Memory (RAM)

• Pin out compatible to the PIC16C72/72A and PIC16F872

• Interrupt capability

• Eight level deep hardware stack

• Direct, Indirect and Relative Addressing modes

Peripheral Features

• Timer0: 8-bit timer/counter with 8-bit preclear

• Timer1: 16-bit timer/counter with preclear, can be incremented during SLEEP via external crystal/clock

• Timer2: 8-bit timer/counter with 8-bit period register, preclear and postscores

• Capture, Compare, PWM(CCP) module

- Capture is 16-bit, max resolution is 12.5 ns

- Compare is 16-bit, max resolution is 200 ns

- PWM max resolution is 10-bit

• 8-bit, 5-channel Analog-to-Digital converter

• Synchronous Serial Port (SSP) with SPI (Master mode) and I2C (Slave)

• Heat sink/Source Current: 25 mA
• Brown-out detection circuitry for Brown-out Reset (BOR)

CMOS Technology:

• Low power, high speed CMOS FLASH technology

• Fully static design

• Wide operating voltage range: 2.0V to 5.5V

• Industrial temperature range

• Low power consumption:

- < 0.6 mA typical @ 3V, 4 MHz

- 20 μA typical @ 3V, 32 kHz

- < 1 μA typical standby current

Following are the major blocks of PIC Microcontroller.

Program memory (FLASH) is used for storing a written program. Since memory made in FLASH technology can be programmed and cleared more than once, it makes this microcontroller suitable for device development. EEPROM - data memory that needs to be saved when there is no supply. It is usually used for storing important data that must not be lost if power supply suddenly stops. For instance, one such data is an assigned temperature in temperature regulators. If during a loss of power supply this data was lost, we would have to make the adjustment once again upon return of supply. Thus our device looses on self-reliance.

RAM- Data memory used by a program during its execution in RAM are stored all inter-results or temporary data during run-time. PORTS are physical connections between the microcontroller and the outside world. PIC16F72 has 22 I/O. FREE-RUN TIMER is an 8-bit register inside a microcontroller that works independently of the program. On every fourth clock of the oscillator it increments its value until it reaches the maximum (255), and then it starts counting over again from zero. As we know the exact timing between each two increments of the timer contents, timer can be used for measuring time which is very useful with some devices.

Crystal oscillator:

The crystal oscillator speed that can be connected to the PIC microcontroller range from DC to 20Mhz. Using the CCS C compiler normally 20Mhz oscillator will be used and the price is very cheap. The 20 MHz crystal oscillator should be connected with about 22pF capacitor. Please refer to my circuit schematic. There are 5 input/output ports on PIC microcontroller namely port A, port B, port C, port D and port E. Each port has different function. Most of them can be used as I/O port.

Clock / instruction cycle

Clock is microcontroller's main starter, and is obtained from an external component called an "oscillator". If we want to compare a microcontroller with a time clock, our "clock" would then be a ticking sound we hear from the time clock. In that case, oscillator could be compared to a spring that is wound so time clock can run. Also, force used to wind the time clock can be compared to an electrical supply. Clock from the oscillator enters a microcontroller via OSC1 pin where internal circuit of a microcontroller divides the clock into four even clocks Q1, Q2, Q3, and Q4 which do not overlap. These four clocks make up one instruction cycle (also called machine cycle) during which one instruction is executed. Execution of instruction starts by calling an instruction that is next in string. Instruction is called from program memory on every Q1 and is written in instruction register on Q4. Decoding and execution of instruction are done between the next Q1 and Q4 cycles. On the following diagram we can see the relationship between instruction cycle and clock of the oscillator (OSC1) as well as that of internal clocks Q1-Q4. Program counter (PC) holds information about the address of the next instruction.

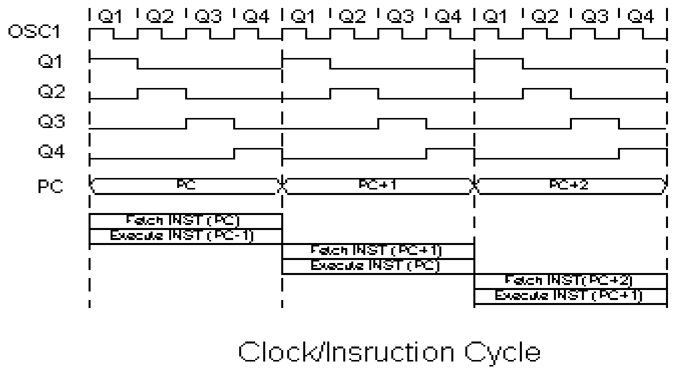
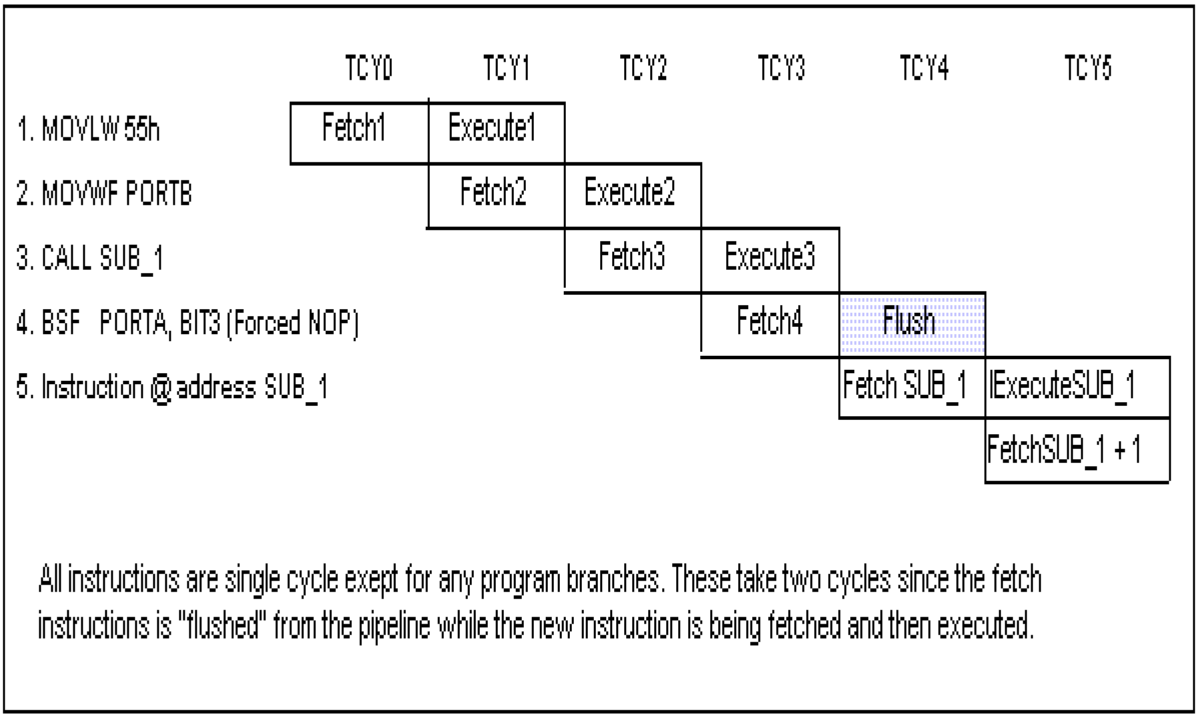


Fig: 2.2 – Clock Instruction Cycle

Pipelining

Instruction cycle consists of cycles Q1, Q2, Q3 and Q4. Cycles of calling and executing instructions are connected in such a way that in order to make a call, one instruction cycle is needed, and one more is needed for decoding and execution. However, due to pipelining, each instruction is effectively executed in one cycle. If instruction causes a change on program counter, and PC doesn't point to the following but to some other address (which can be the case with jumps or with calling subprograms), two cycles are needed for executing an instruction. This is so because instruction must be processed again, but this time from the right address. Cycle of calling begins with Q1 clock, by writing into instruction register (IR). Decoding and executing begins with Q2, Q3 and Q4 clocks.



Instruction Pipeline Flow

***Fig: 2.3*- Instruction Pipeline Flow**

Pin description

PIC16F72 has a total of 28 pins. It is most frequently found in a DIP28 type of case but can also be found in SMD case which is smaller from a DIP. DIP is an abbreviation for Dual In Package. SMD is an abbreviation for Surface Mount Devices suggesting that holes for pins to go through when mounting aren't necessary in soldering this type of a component.

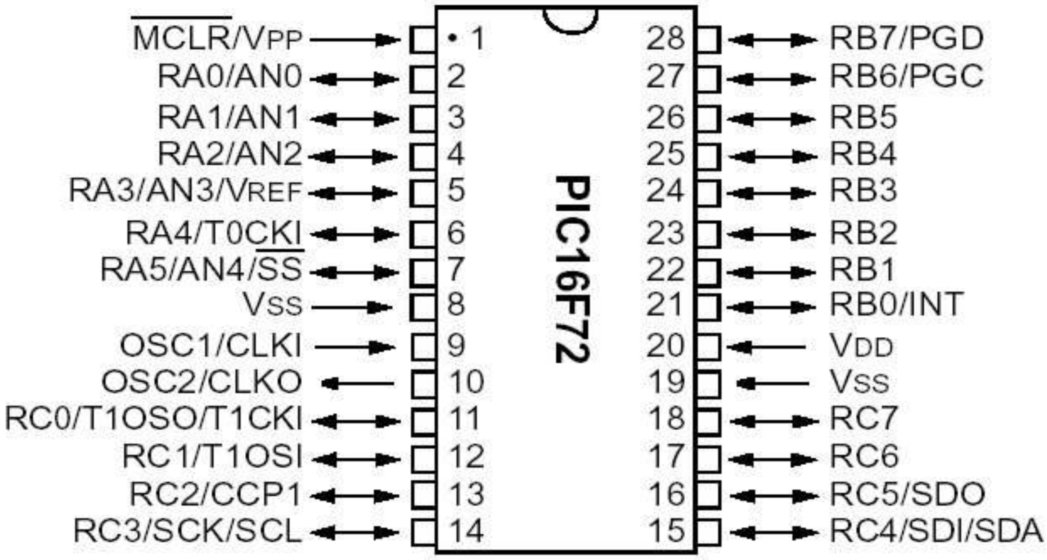


***Fig*: 2.4- Pin Description**

Pins on PIC16F72 microcontroller have the following meaning:

There are 28 pins on PIC16F72. Most of them can be used as an IO pin. Others are already for specific functions. These are the pin functions.

1. MCLR – to reset the PIC
2. RA0 – port A pin 0
3. RA1 – port A pin 1
4. RA2 – port A pin 2
5. RA3 – port A pin 3
6. RA4 – port A pin 4
7. RA5 – port A pin 5
8. VSS – ground
9. OSC1 – connect to oscillator
10. OSC2 – connect to oscillator
11. RC0 – port C pin 0 VDD – power supply
12. RC1 – port C pin 1
13. RC2 – port C pin 2
14. RC3 – port C pin 3
15. RC4 - port C pin 4
16. RC5 - port C pin 5

17.	RC6	-	port	C	pin	6
18.	RC7	-	port	C	pin	7
19.	VSS	-	ground			
20.	VDD	–	power supply			
21.	RB0	-	port	B	pin	0
22.	RB1	-	port	B	pin	1
23.	RB2	-	port	B	pin	2
24.	RB3	-	port	B	pin	3
25.	RB4	-	port	B	pin	4
26.	RB5	-	port	B	pin	5
27.	RB6	-	port	B	pin	6
28.	RB7	-	port	B	pin	7

By utilizing all of this pin so many application can be done such as:

1. LCD – connect to Port B pin.
2. LED – connect to any pin declared as output.
3. Relay and Motor - connect to any pin declared as output.
4. External EEPROM – connect to I2C interface pin – RC3 and RC4 (SCL and SDA)
5. LDR, Potentiometer and sensor – connect to analogue input pin such as RA0. 6. GSM modem dial up modem – connect to RC6 and RC7 – the serial communication interface using RS232 protocol. For more detail function for each specific pin please refer to the device datasheet from Microchip.

Ports

Term "port" refers to a group of pins on a microcontroller which can be accessed simultaneously, or on which we can set the desired combination of zeros and ones, or read from them an existing status. Physically, port is a register inside a microcontroller which is connected by wires to the pins of a microcontroller. Ports represent physical connection of Central Processing Unit with an outside world. Microcontroller uses them in order to monitor or control other components or devices. Due to functionality, some pins have twofold roles like PA4/TOCKI for instance, which is in the same time the fourth bit of port A and an external input for free-run counter. Selection of one of these two pin functions is done in one of the configuration registers. An illustration of this is the fifth bit T0CS in OPTION register. By selecting one of the functions the other one is disabled. All port pins can be designated as input or output, according to the needs of a device that's being developed. In order to define a pin as input or output pin, the right combination of zeros and ones must be written in TRIS register. If the appropriate bit of TRIS register contains logical "1", then that pin is an input pin, and if the opposite is true, it's an output pin. Every port has its proper TRIS register. Thus, port A has TRISA, and port B has TRISB. Pin direction can be changed during the course of work which is particularly fitting for one-line communication where data flow constantly changes direction. PORTA and PORTB state registers are located in bank 0, while TRISA and TRISB pin direction registers are located in bank 1.

PORTB and TRISB

PORTB have adjoined 8 pins. The appropriate register for data direction is TRISB. Setting a bit in TRISB register defines the corresponding port pin as input, and resetting a bit in TRISB register defines the corresponding port pin as output.

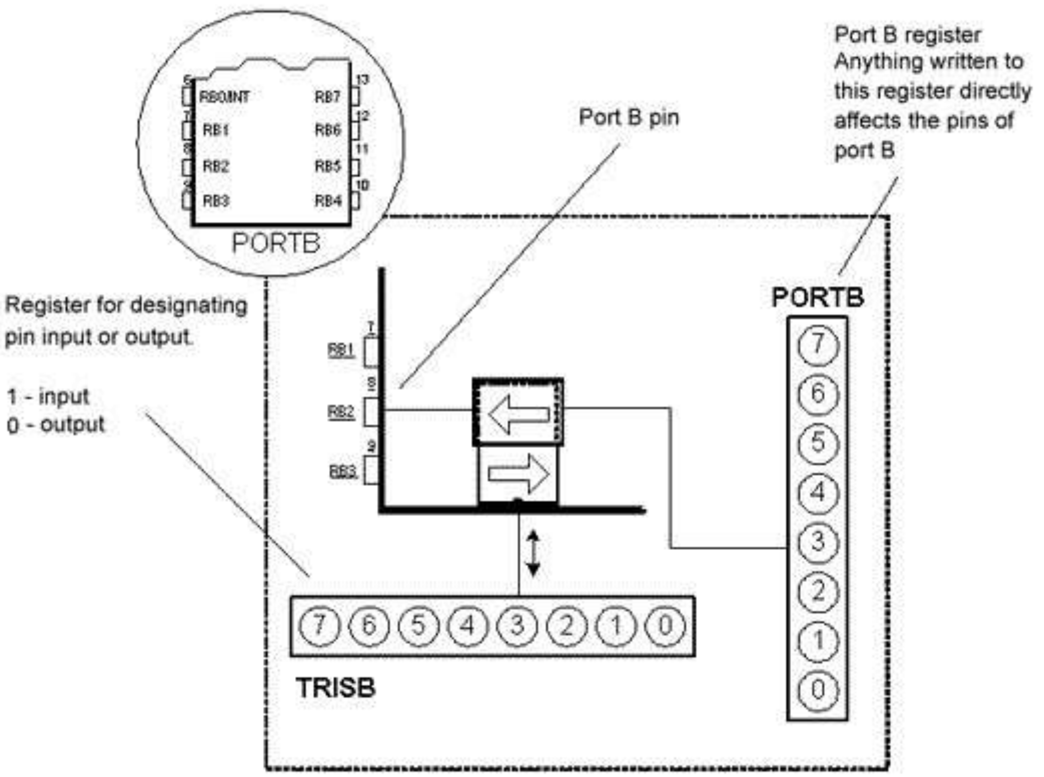


Fig: 2.5- **PORTB and TRISB**

Each PORTB pin has a weak internal pull-up resistor (resistor which defines a line to logic one) which can be activated by resetting the seventh bit RBPU in OPTION register. These 'pull-up' resistors are automatically being turned off when port pin is configured as an output. When a microcontroller is started, pull-ups are disabled.

Four pins PORTB, RB7:RB4 can cause an interrupt which occurs when their status changes from logical one into logical zero and opposite. Only pins configured as input can cause this interrupt to occur (if any RB7:RB4 pin is configured as an output, an interrupt won't be generated at the change of status.) This interrupt option along with internal pull-up resistors makes it easier to solve common problems we find in practice like for instance that of matrix keyboard. If rows on the keyboard are connected to these pins, each push on a key will then cause an interrupt. A microcontroller will determine which key is at hand while processing an interrupt It is not recommended to refer to port B at the same time that interrupt is being processed.

PORTA and TRISA

PORTA have 5 adjoining pins. The corresponding register for data direction is TRISA at address 85h. Like with port B, setting a bit in TRISA register defines also the corresponding port pin as input, and clearing a bit in TRISA register defines the corresponding port pin as output.

It is important to note that PORTA pin RA4 can be input only. On that pin is also situated an external input for timer TMR0. Whether RA4 will be a standard input or an input for a counter depends on T0CS bit (TMR0 Clock Source Select bit). This pin enables the timer TMR0 to increment either from internal oscillator or via external impulses on RA4/T0CKI pin.

Example shows how pins 0, 1, 2, 3, and 4 are designated input, and pins 5, 6, and 7 outputs. After this, it is possible to read the pins RA2, RA3, RA4, and to set logical zero or one to pins RA0 and RA1.

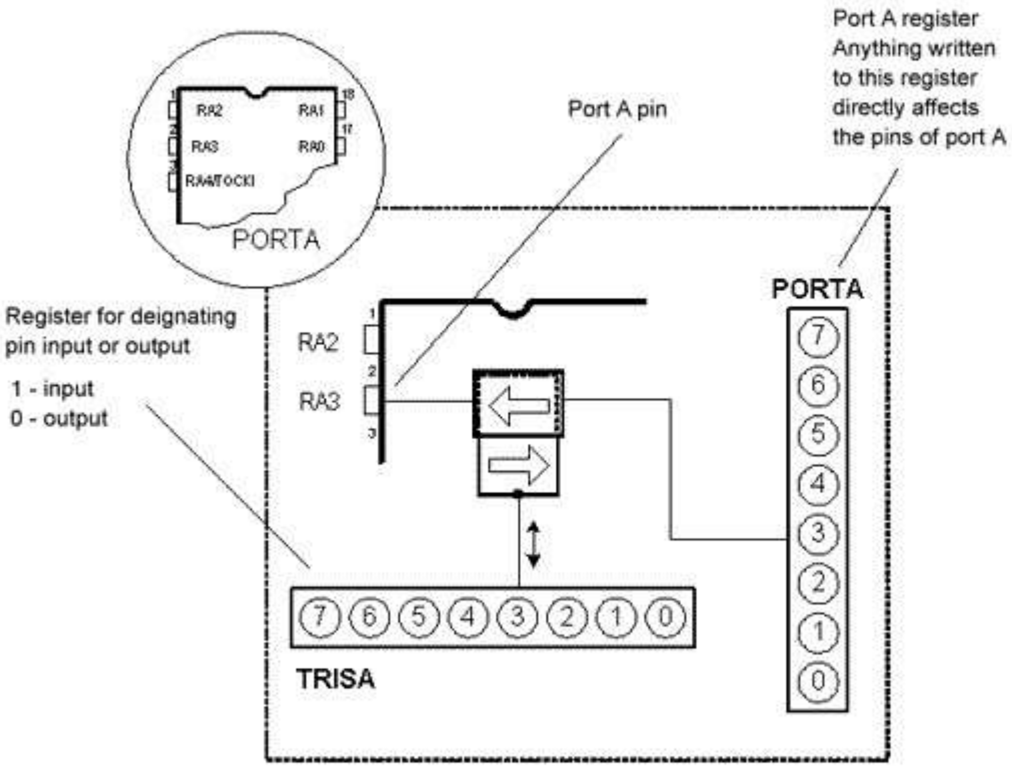


Fig: 2.6- PORTA and TRISA

Memory organization

PIC16F72 has two separate memory blocks, one for data and the other for program. EEPROM memory with GPR and SFR registers in RAM memory make up the data block, while FLASH memory makes up the program block.

Program memory

Program memory has been carried out in FLASH technology which makes it possible to program a microcontroller many times before it's installed into a device, and even after its instalment if eventual changes in program or process parameters should occur. The size of program memory is 1024 locations with 14 bits width where locations zero and four are reserved for reset and interrupt vector.

Data memory

Data memory consists of EEPROM and RAM memories. EEPROM memory consists of 256 eight bit locations whose contents are not lost during loosing of power supply. EEPROM is not directly addressable, but is accessed indirectly through EEADR and EEDATA registers. As EEPROM memory usually serves for storing important parameters (for example, of a given temperature in temperature regulators) , there is a strict procedure for writing in EEPROM which must be followed in order to avoid accidental writing.

RAM memory for data occupies space on a memory map from location 0x0C to 0x4F which comes to 68 locations. Locations of RAM memory are also called GPR registers which is an abbreviation for *General Purpose Registers*. GPR registers can be accessed regardless of which bank is selected at the moment.

Applications

PIC16F72 perfectly fits many uses, from automotive industries and controlling home appliances to industrial instruments, remote sensors, electrical door locks and safety devices. It is also ideal for smart cards as well as for battery supplied devices because of its low consumption.

EEPROM memory makes it easier to apply microcontrollers to devices where permanent storage of various parameters is needed (codes for transmitters, motor speed, receiver frequencies, etc.). Low cost, low consumption, easy handling and flexibility make PIC16F72 applicable even in areas where microcontrollers had not previously been considered (example: timer functions, interface replacement in larger systems, coprocessor applications, etc.).

In System Programmability of this chip (along with using only two pins in data transfer) makes possible the flexibility of a product, after assembling and testing have been completed. This capability can be used to create assembly-line production, to store calibration data available only after final testing, or it can be used to improve programs on finished products.

2.3 REGULATED POWER SUPPLY

2.3.1 Introduction:

Power supply is a supply of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.A power supply may include a power distribution system as well as primary or secondary sources of energy such as

- Conversion of one form of electrical power to another desired form and voltage, typically involving converting AC line voltage to a well-regulated lower voltage DC for electronic devices. Low voltage, low power DC power supply units are commonly integrated with the devices they supply, such as computers and household electronics.

- Batteries.
- Chemical fuel cells and other forms of energy storage systems.
- Solar power.
- Generators or alternators.

2.3.2 Block Diagram:

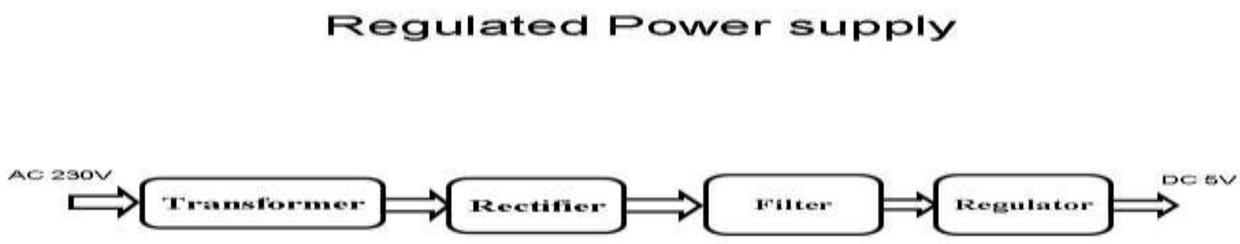


Fig: 2.7 Regulated Power Supply

The basic circuit diagram of a regulated power supply (DC O/P) with led connected as load is shown in fig: 3.3.3.

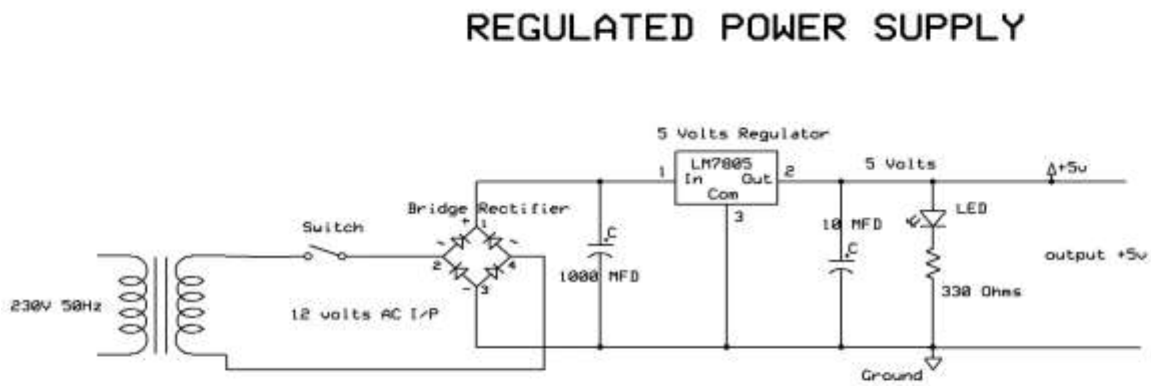


Fig 2.7.1 Circuit diagram of Regulated Power Supply with Led connection

components mainly used in above figure are

- 230V AC MAINS
- TRANSFORMER
- BRIDGE RECTIFIER(DIODES)

- CAPACITOR
- VOLTAGE REGULATOR(IC 7805)
- RESISTOR
- LED(LIGHT EMITTING DIODE)

The detailed explanation of each and every component mentioned above is as follows:

Transformation: The process of transforming energy from one device to another is called transformation. For transforming energy we use transformers.

Transformers:

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled conductors without changing its frequency. A varying current in the first or primary winding creates a varying magnetic flux in the transformer's core, and thus a varying magnetic field through the secondary winding. This varying magnetic field induces a varying electromotive force (EMF) or "voltage" in the secondary winding. This effect is called mutual induction. If a load is connected to the secondary, an electric current will flow in the secondary winding and electrical energy will be transferred from the primary circuit through the transformer to the load. This field is made up from lines of force and has the same shape as a bar magnet. If the current is increased, the lines of force move outwards from the coil. If the current is reduced, the lines of force move inwards. If another coil is placed adjacent to the first coil then, as the field moves out or in, the moving lines of force will "cut" the turns of the second coil. As it does this, a voltage is induced in the second coil. With the 50 Hz AC mains supply, this will happen 50 times a second. This is called MUTUAL INDUCTION and forms the basis of the transformer.

The input coil is called the PRIMARY WINDING; the output coil is the SECONDARY WINDING. Fig: 3.3.4 shows step-down transformer.

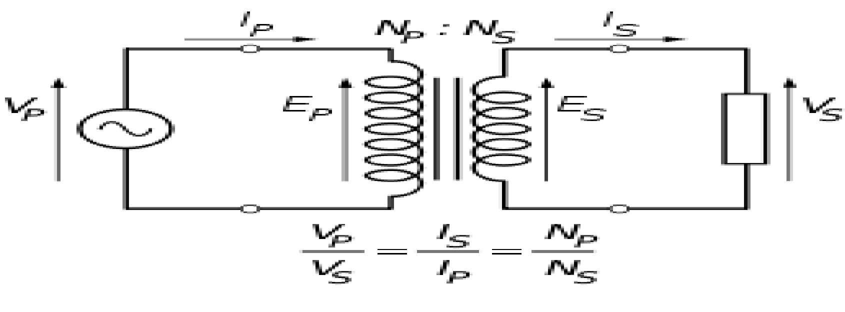


Fig 2.8: Step-Down Transformer

The voltage induced in the secondary is determined by the TURNS RATIO.

$$\frac{\textbf{primary voltage}}{\textbf{secondary voltage}} = \frac{\textbf{number of primary turns}}{\textbf{number of secondary turns}}$$

For example, if the secondary has half the primary turns; the secondary will have half the primary voltage.

Another example is if the primary has 5000 turns and the secondary has 500 turns, then the turn's ratio is 10:1.

If the primary voltage is 240 volts then the secondary voltage will be x 10 smaller = 24 volts. Assuming a perfect transformer, the power provided by the primary must equal the power taken by a load on the secondary. If a 24-watt lamp is connected across a 24 volt secondary, then the primary must supply 24 watts.

To aid magnetic coupling between primary and secondary, the coils are wound on a metal CORE. Since the primary would induce power, called EDDY CURRENTS, into this core, the core is LAMINATED. This means that it is made up from metal sheets insulated from each other. Transformers to work at higher frequencies have an iron dust core or no core at all.

Note that the transformer only works on AC, which has a constantly changing current and moving field. DC has a steady current and therefore a steady field and there would be no induction.

Some transformers have an electrostatic screen between primary and secondary. This is to prevent some types of interference being fed from the equipment down into the mains supply, or in the other direction. Transformers are sometimes used for IMPEDANCE MATCHING.

We can use the transformers as step up or step down.

Step Up transformer:

In case of step up transformer, primary windings are every less compared to secondary winding. Because of having more turns secondary winding accepts more energy, and it releases more voltage at the output side.

Step down transformer:

In case of step down transformer, Primary winding induces more flux than the secondary winding, and secondary winding is having less number of turns because of that it accepts less number of flux, and releases less amount of voltage.

RECTIFICATION:

The process of converting an alternating current to a pulsating direct current is called as rectification. For rectification purpose we use rectifiers.

Rectifiers:

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC), a process known as rectification. Rectifiers have many uses including as components of power supplies and as detectors of radio signals. Rectifiers may be made of solid-state diodes, vacuum tube diodes, mercury arc valves, and other components. A device that it can perform the opposite function (converting DC to AC) is known as an inverter. When only one diode is used to rectify AC (by blocking the negative or positive portion of the waveform), the difference between the term diode and the term rectifier is merely one of usage, i.e., the term rectifier describes a diode that is being used to convert AC to DC. Almost all rectifiers comprise a number of diodes in a specific arrangement for more efficiently converting AC to DC than is possible with only one diode. Before the development of silicon semiconductor rectifiers, vacuum tube diodes and copper (I) oxide or selenium rectifier stacks were used.

Bridge full wave rectifier:

The Bridge rectifier circuit is shown in fig: 3.3.7, which converts an ac voltage to dc voltage using both half cycles of the input ac voltage. The Bridge rectifier circuit is shown in the figure. The circuit has four diodes connected to form a bridge. The ac input voltage is applied to the diagonally opposite ends of the bridge. The load resistance is connected between the other two ends of the bridge. For the positive half cycle of the input ac voltage, diodes D1 and D3 conduct, whereas diodes D2 and D4 remain in the OFF state. The conducting diodes will be in series with the load resistance R_L and hence the load current flows through R_L. For the negative half cycle of the input ac voltage, diodes D2 and D4 conduct whereas, D1 and D3 remain OFF. The conducting diodes D2 and D4 will be in series with the load resistance R_L and hence the current flows through R_L in the same direction as in the previous half cycle. Thus a bidirectional wave is converted into a unidirectional wave.

Input Output

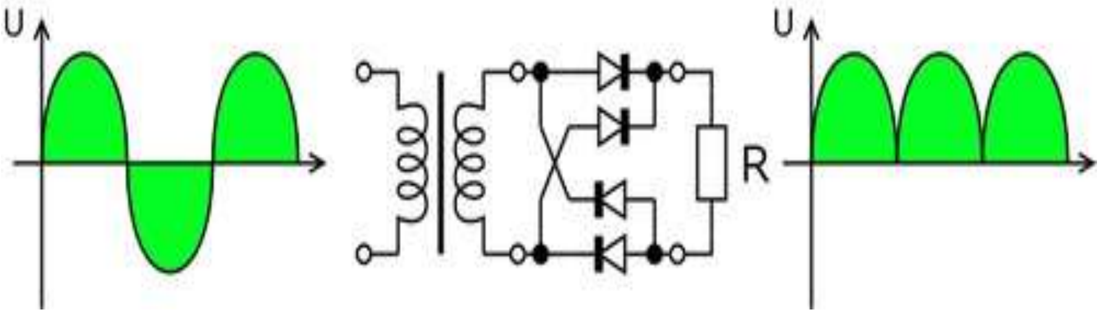


Fig 2.9: Bridge rectifier: a full-wave rectifier using 4 diodes

DB107: Now -a -days Bridge rectifier is available in IC with a number of DB107. In our project we are using an IC in place of bridge rectifier. The picture of DB 107 is shown in fig: 3.3.8.

Features:

- Good for automation insertion
- Surge overload rating - 30 amperes peak
- Ideal for printed circuit board
- Reliable low cost construction utilizing molded
- Glass passivated device
- Polarity symbols moulded on body
- Mounting position: Any
- Weight: 1.0 gram

Fig 2.10: DB107

Filtration:

The process of converting a pulsating direct current to a pure direct current using filters is called as filtration.

Filters:

Electronic filters are electronic circuits, which perform signal-processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones.

Introduction to Capacitors:

The Capacitor or sometimes referred to as a Condenser is a passive device, and one which stores energy in the form of an electrostatic field which produces a potential (static voltage) across its plates. In its basic form a capacitor consists of two parallel conductive plates that are not connected but are electrically separated either by air or by an insulating material called the **Dielectric**. When a voltage is applied to these plates, a current flows charging up the plates with electrons giving one plate a positive charge and the other plate an equal and opposite negative charge This flow of electrons to the plates is known as the Charging Current and continues to flow until the voltage across the plates (and hence the capacitor) is equal to the applied voltage Vcc. At this point the capacitor is said to be fully charged and this is illustrated below. The construction of capacitor and an electrolytic capacitor.

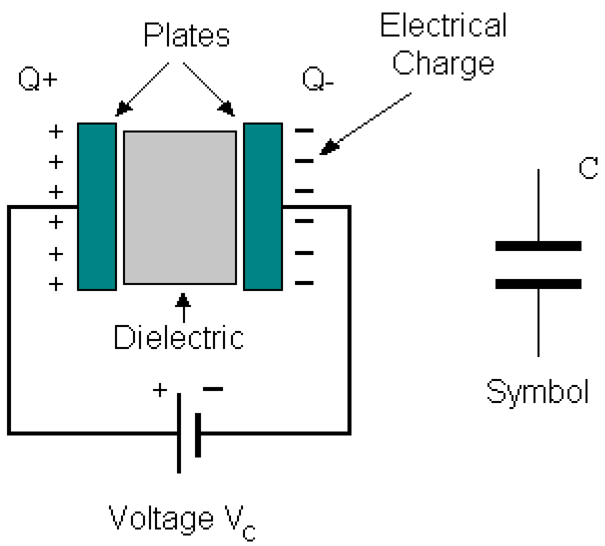


Fig 2.11: Construction Of a Capacitor

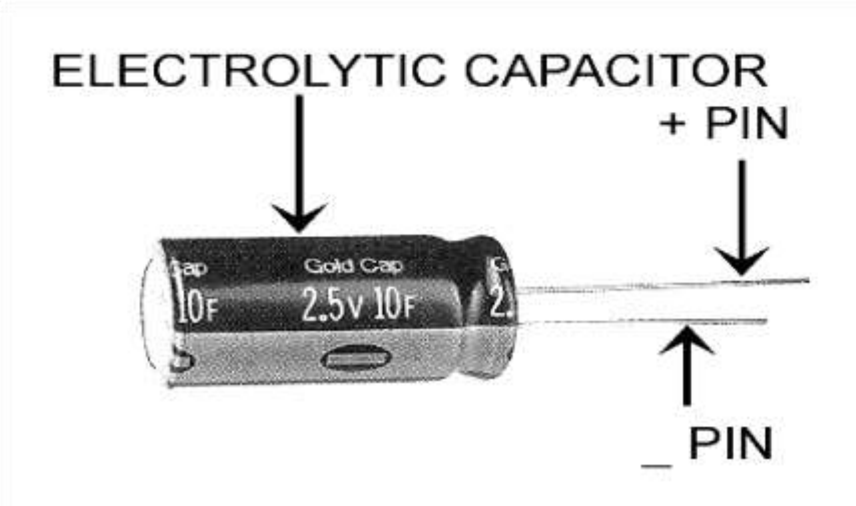


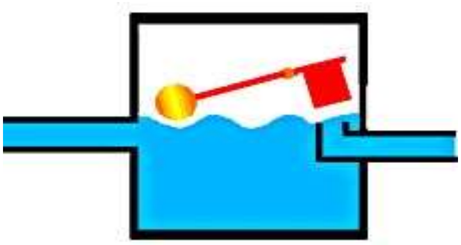
Fig 2.12: Electrolytic Capaticor Units of Capacitance:

Microfarad (μF) 1μF = 1/1,000,000 = 0.000001 = 10^{-6} F

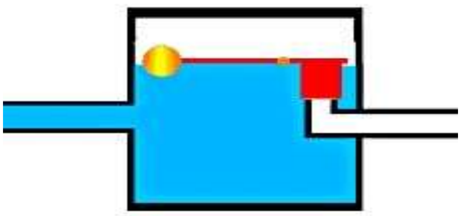
Nanofarad (nF) 1nF = 1/1,000,000,000 = 0.000000001 = 10^{-9} F

Pico farad (pF) 1pF = 1/1,000,000,000,000 = 0.000000000001 = 10^{-12} F **Operation of Capacitor:**

Think of water flowing through a pipe. If we imagine a capacitor as being a storage tank with an inlet and an outlet pipe, it is possible to show approximately how an electronic capacitor works. First, let's consider the case of a "coupling capacitor" where the capacitor is used to connect a signal from one part of a circuit to another but without allowing any direct current to flow.



If the current flow is alternating between zero and a maximum, our "storage tank" capacitor will allow the current waves to pass through.



However, if there is a steady current, only the initial short burst will flow until the "floating ball valve" closes and stops further flow.

So a coupling capacitor allows "alternating current" to pass through because the ball valve doesn't get a chance to close as the waves go up and down. However, a steady current quickly fills the tank so that all flow stops. A capacitor will pass alternating current but (apart from an initial surge) it will not pass d.c.

Where a capacitor is used to decouple a circuit, the effect is to "smooth out ripples". Any ripples, waves or PIC microcontroller based pulses of current are passed to ground while d.c. Flows smoothly.

Regulation:

The process of converting a varying voltage to a constant regulated voltage is called as regulation. For the process of regulation we use voltage regulators.

Voltage Regulator:

A voltage regulator (also called a 'regulator') with only three terminals appears to be a simple device, but it is in fact a very complex integrated circuit. It converts a varying input voltage into a constant 'regulated' output voltage. Voltage Regulators are available in a variety of outputs like 5V, 6V, 9V, 12V and 15V. The LM78XX series of voltage regulators are designed for positive input. For applications requiring negative input, the LM79XX series is used. Using a pair of 'voltage-divider' resistors can increase the output voltage of a regulator circuit.

It is not possible to obtain a voltage lower than the stated rating. You cannot use a 12V regulator to make a 5V power supply. Voltage regulators are very robust. These can withstand over-current draw due to short circuits and also over-heating. In both cases, the regulator will cut off before any damage occurs. The only way to destroy a regulator is to apply reverse voltage to its input. Reverse polarity destroys the regulator almost instantly. Fig: 3.3.11 shows voltage regulator.

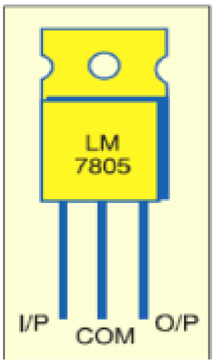


Fig 2.13: Voltage Regulator

Resistors:

A resistor is a two-terminal electronic component that produces a voltage across its terminals that is proportional to the electric current passing through it in accordance with Ohm's law:

V = IR

Resistors are elements of electrical networks and electronic circuits and are ubiquitous in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel/chrome). The primary characteristics of a resistor are the resistance, the tolerance, maximum working voltage and the power rating. Other characteristics include temperature coefficient, noise, and inductance. Less well-known is critical resistance, the value below which power dissipation limits the maximum permitted current flow, and above which the limit is applied voltage. Critical resistance is determined by the design, materials and dimensions of the resistor.

Resistors can be made to control the flow of current, to work as Voltage dividers, to dissipate power and it can shape electrical waves when used in combination of other components. Basic unit is ohms.

Theory of operation:

Ohm's law:

The behaviour of an ideal resistor is dictated by the relationship specified in Ohm's law:

$$V = IR$$

Ohm's law states that the voltage (V) across a resistor is proportional to the current (I) through it where the constant of proportionality is the resistance (R).

Power dissipation: The power dissipated by a resistor (or the equivalent resistance of a resistor network) is calculated using the following:

$$P = I^2 R = IV = \frac{V^2}{R}$$

Fig 2.14: Resistor

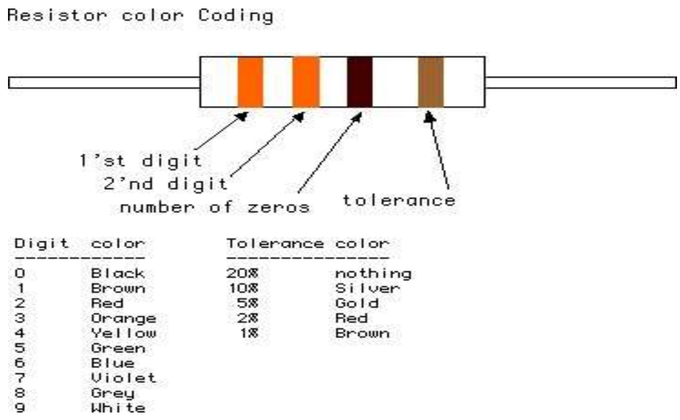


Fig 2.15: Color Bands In Resistor

2.4. LED:

A light-emitting diode (LED) is a semiconductor light source. LED's are used as indicator lamps in many devices, and are increasingly used for lighting. Introduced as a practical electronic component in 1962, early LED's emitted low-intensity red light, but modern versions are available across the visible, ultraviolet and infrared wavelengths, with very high brightness. The internal structure and parts of a led are shown in figures 3.4.1 and 3.4.2 respectively.

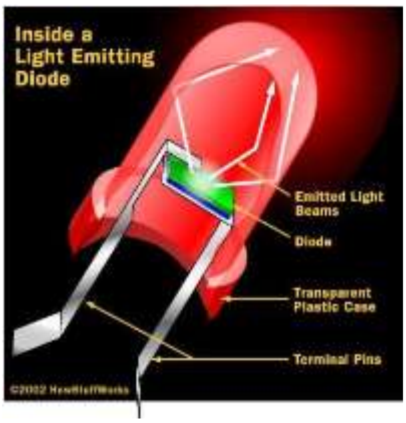


Fig 2.16: Inside a LED

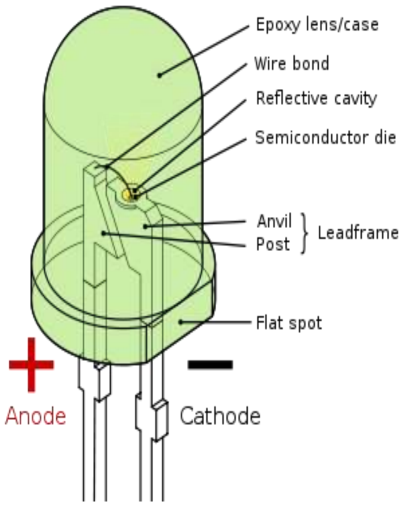


Fig 2.17: Parts of a LED

Working:

The structure of the LED light is completely different than that of the light bulb. Amazingly, the LED has a simple and strong structure. The light-emitting semiconductor material is what determines the LED's colour. The LED is based on the semiconductor diode.

When a diode is forward biased (switched on), electrons are able to recombine with holes within the device, releasing energy in the form of photons. This effect is called electroluminescence and the colour of the light (corresponding to the energy of the photon) is determined by the energy gap of the semiconductor. An LED is usually small in area (less than 1 mm^2), and integrated optical components are used to shape its radiation pattern and assist in reflection. LED's present many advantages over incandescent light sources including lower energy consumption, longer lifetime, improved robustness, smaller size, faster switching, and greater durability and reliability. However, they are relatively expensive and require more precise current and heat management than traditional light sources. Current LED products for general lighting are more expensive to buy than fluorescent lamp sources of comparable output. They also enjoy use in applications as diverse as replacements for traditional light sources in automotive lighting (particularly indicators) and in traffic signals. The compact size of LED's has allowed new text and video displays and sensors to be developed, while their high switching rates are useful in advanced communications technology. The electrical symbol and polarities of led are shown in fig: 3.4.3.

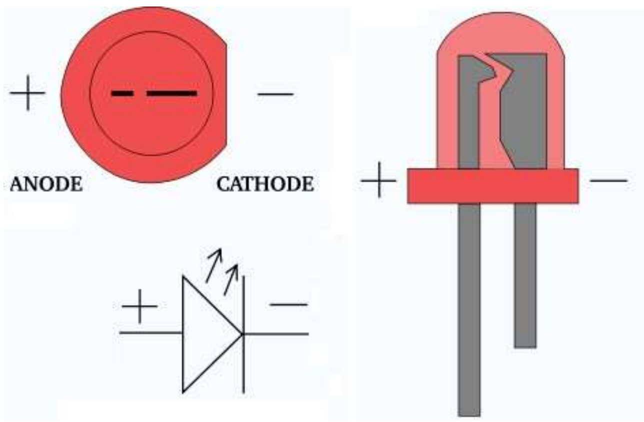


Fig 2.18: Electrical Symbol & Polarities of LED

LED lights have a variety of advantages over other light sources:

- High-levels of brightness and intensity
- High-efficiency
- Low-voltage and current requirements
- Low radiated heat
- High reliability (resistant to shock and vibration)
- No UV Rays
- Long source life
- Can be easily controlled and programmed

Applications of LED fall into three major categories:

- Visual signal application where the light goes more or less directly from the LED to the human eye, to convey a message or meaning.
- Illumination where LED light is reflected from object to give visual response of these objects.
- Generate light for measuring and interacting with processes that do not involve the human visual system.

2.5 Copper coils

Fig: 2.19 Copper coil

An *electromagnetic coil* (or simply a "coil") is formed when a conductor (usually an insulated solid copper wire) is wound around a core or form to create an inductor or electromagnet. One loop of wire is usually referred to as a *turn*, and a coil consists of one or more turns. For use in an electronic circuit, electrical connection terminals called taps are often connected to a coil. Coils are often coated with varnish or wrapped with insulating tape to provide additional insulation and secure them in place. A completed coil assembly with taps is often called a *winding*. A transformer is an electromagnetic device that has a *primary winding* and a *secondary winding* that transfers energy from one electrical circuit to another by inductive coupling without moving parts. The term *tickler coil* usually refers to a feedback coil, which is often the third coil placed in relation to a primary coil and secondary coil.

A coil tap is a wiring feature found on some electrical transformers, inductors and coil pickups, all of which are sets of wire coils. The coil tap(s) are points in a wire coil where a conductive patch has been exposed (usually on a loop of wire that extends out of the main coil body). As self-induction is larger for larger coil diameter the current in a thick wire tries to flow on the inside. The ideal use of copper is achieved by foils. Sometimes this means that a spiral is a better alternative. Multilayer coils have the problem of interlayer capacitance, so when multiple layers are needed the shape needs to be radically changed to a short coil with many layers so that the voltage between consecutive layers is smaller.

5V 2A Wireless Power Supply Transmitter Receiver Charging Coil Module:

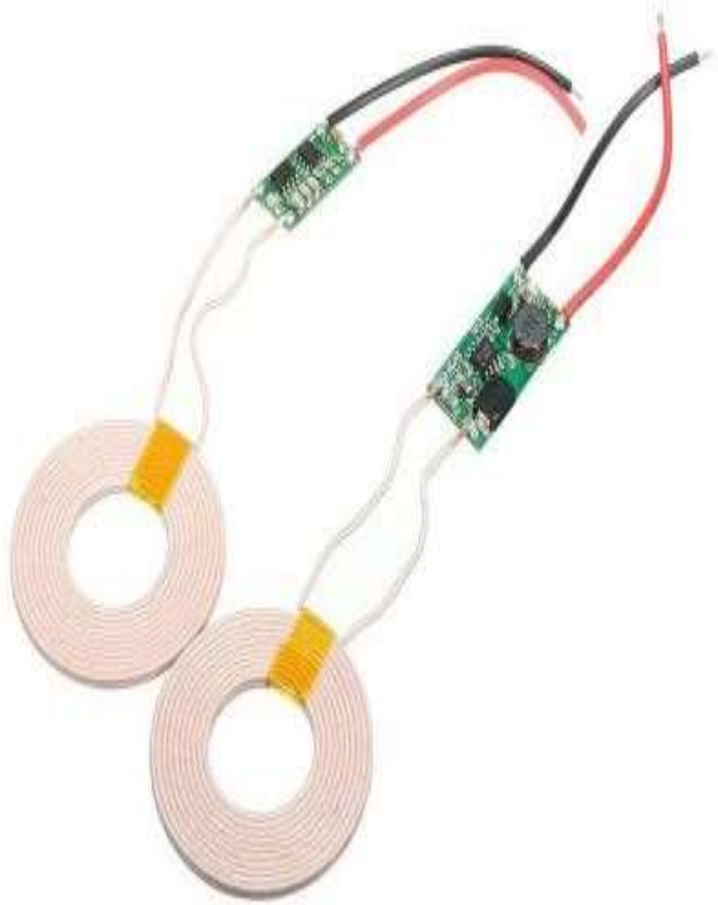


Fig 2.20: wireless power transmitter-receiver coils

Transmitting Part:

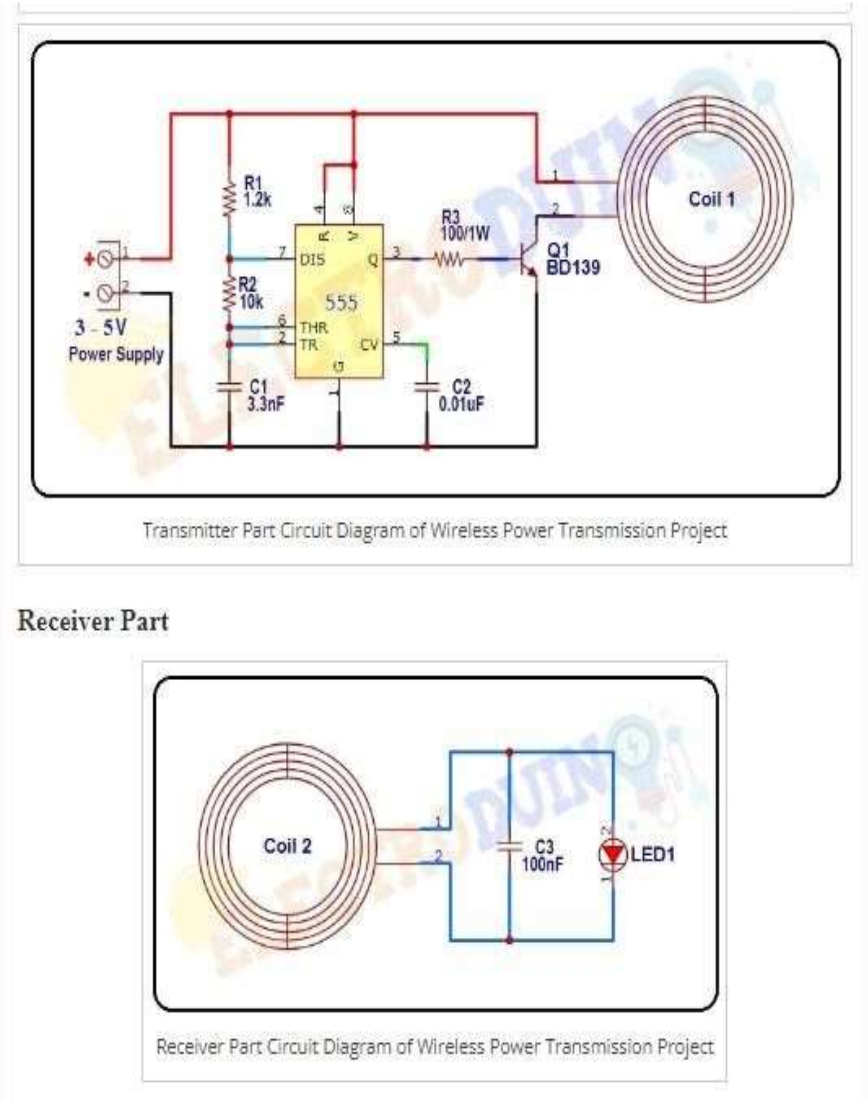


Fig: 2.21 Transmitter circuit and Receiver circuit

The 5V 2A Large Current Wireless Charger Module Transmitter Receiver Charging Coil Module is for a variety of small electronic products, wireless charging, power supply development, and design, with a small size, easy to use, high efficiency, low price characteristics. It is mainly used in mobile electronics products such as for charging the mobile phone wirelessly, game machines, tanks, MP3, MP4, adult products, digital cameras, electric shavers, machine learning, health care, etc. Due to the adoption of a Contactless charging power supply, to complete sealing of the products, waterproof and dust-proof; increase product service life, use more convenient.

Table:2.1 Specifications of Coils

Specifications:	
Model	XKT-412
Input Voltage	12VDC
Output	5V / 1A current
Normal Use Distance	2 ~ 10mm
Operating Current	1.2-2 A
Tx/Rx Coil Dimensions	Diameter 43mm, Inner diameter 20mm
Coil Wire Thickness	2.3mm
The transmitting module size	18mmx8.5mmx15mm
The receiving module size	10mmx25mmx3mm

2.6 Design and Analysis Of 12v Charging Circuit

Rechargeable battery:

A rechargeable battery, storage battery, or accumulator is a type of electrical battery. It comprises one or more electrochemical cells, and is a type of energy accumulator. It is known as a secondary cell because its electrochemical reactions are electrically reversible. Rechargeable batteries come in many different shapes and sizes, ranging from button cells to megawatt systems connected to stabilize an electrical distribution network. Several different combinations of chemicals are commonly used, including: lead–acid, nickel cadmium (NiCd), nickel metal hydride (NiMH), lithium ion (Li-ion), and lithium ion polymer (Li-ion polymer).

Fig: 2.22 Rechargeable Battery

Charging Circuit:

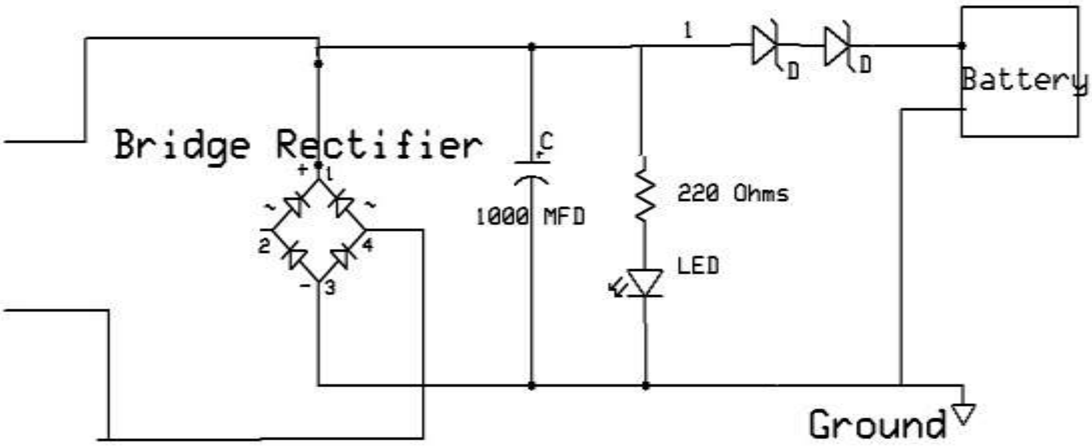


Fig: 2.23 Charging circuit

Above circuit diagram take the input from 12v 5watt Solar panel and, we can see that the 12v DC is being stored into the battery with the help of rectifier and 1000MF capacitor used. We can get 12V,1ahm Steady DC at the output terminal is fed to the rechargeable battery which can be indicated if the LED glows. Here we are using diodes for reverse current protection. Diodes will conduct in forward bias.

.

2.7 Solar Panel:

Photovoltaic Cells: Converting Photons to Electrons

The solar cells that you see on calculators and satellites are also called photovoltaic (PV) cells, which as the name implies (photo meaning "light" and voltaic meaning "electricity"), convert sunlight directly into electricity. A module is a group of cells connected electrically and packaged into a frame (more commonly known as a solar panel), which can then be grouped into larger solar arrays.

Photovoltaic cells are made of special materials called semiconductors such as silicon, which is currently used most commonly. Basically, when light strikes the cell, a certain portion of it is absorbed within the semiconductor material. This means that the energy of the absorbed light is transferred to the semiconductor. The energy knocks electrons loose, allowing them to flow freely.

PV cells also all have one or more electric field that acts to force electrons freed by light absorption to flow in a certain direction. This flow of electrons is a current, and by placing metal contacts on the top and bottom of the PV cell, we can draw that current off for external use, say, to power a calculator. This current, together with the cell's voltage (which is a result of its built-in electric field or fields), defines the power (or wattage) that the solar cell can produce.

Solar Panel Setup:

The use of batteries requires the installation of another component called a **charge controller**. Batteries last a lot longer if they aren't overcharged or drained too much. That's what a charge controller does. Once the batteries are fully charged, the charge controller doesn't let current from the PV modules continue to flow into them. Similarly, once the batteries have been drained to a certain predetermined level, controlled by measuring battery voltage, many charge controllers will not allow more current to be drained from the batteries until they have been recharged. The use of a charge controller is essential for long battery life. The other problem besides energy storage is that the electricity generated by your solar panels, and extracted from your batteries if you choose to use them, is not in the form that's supplied by your utility or used by the electrical appliances in your house. The electricity generated by a solar system is direct current, so you'll need an **inverter** to convert it into alternating current. Most large inverters will allow you to automatically control how your system works. Some PV modules, called **AC modules**, actually have an inverter already built into each module, eliminating the need for a large, central inverter, and simplifying wiring issues. Throw in the mounting hardware, wiring, junction boxes, grounding equipment, over current protection, DC and AC disconnects and other accessories, and you have yourself a system. You must follow electrical codes (there's a section in the National Electrical Code just for PV), and it's highly recommended that a licensed electrician who has experience with PV systems do the installation. Once installed, a PV system requires very little maintenance (especially if no batteries are used), and will provide electricity cleanly and quietly for 20 years or more.

2.7.1 Solar energy:

Solar energy, radiant light and heat from the sun, has been harnessed by humans since ancient times using a range of ever-evolving technologies. Solar radiation, along with secondary solar-powered resources such as wind and wave power, hydroelectricity and biomass, account for most of the available renewable energy on earth Only a minuscule fraction of the available solar energy is used.

Solar powered electrical generation relies on heat engines and photovoltaic. Solar energy's uses are limited only by human ingenuity. A partial list of solar applications includes space heating and cooling through solar architecture, potable water via distillation and disinfection, day lighting, solar hot water, solar cooking, and high temperature process heat for industrial purposes. To harvest the solar energy, the most common way is to use solar panels.

Solar technologies are broadly characterized as either passive solar or active solar depending on the way they capture, convert and distribute solar energy. Active solar techniques include the use of photovoltaic panels and solar thermal collectors to harness the energy. Passive solar techniques include orienting a building to the Sun, selecting materials with favourable thermal mass or light dispersing properties, and designing spaces that naturally circulate air.

The Earth receives 174 petawatts (PW) of incoming solar radiation (insolation) at the upper atmosphere. Approximately 30% is reflected back to space while the rest is absorbed by clouds, oceans and land masses. The spectrum of solar light at the Earth's surface is mostly spread across the visible and near-infrared ranges with a small part in the near-ultraviolet.

Earth's land surface, oceans and atmosphere absorb solar radiation, and this raises their temperature. Warm air containing evaporated water from the oceans rises, causing atmospheric circulation or convection. When the air reaches a high altitude, where the temperature is low, water vapor condenses into clouds, which rain onto the Earth's surface, completing the water cycle. The latent heat of water condensation amplifies convection, producing atmospheric phenomena such as wind, cyclones and anticyclones. Sunlight absorbed by the oceans and land masses keeps the surface at an average temperature of

14 °C. By photosynthesis green plants convert solar energy into chemical energy, which produces food, wood and the biomass from which fossil fuels are derived.

The total solar energy absorbed by Earth's atmosphere, oceans and land masses is approximately 3,850,000 exajoules (EJ) per year.[6] In 2002, this was more energy in one hour than the world used in one year. Photosynthesis captures approximately 3,000 EJ per year in biomass. The amount of solar energy reaching the surface of the planet is so vast that in one year it is about twice as much as will ever be obtained from all of the Earth's non-renewable resources of coal, oil, natural gas, and mined uranium combined.

From the table of resources it would appear that solar, wind or biomass would be sufficient to supply all of our energy needs, however, the increased use of biomass has had a negative effect on global warming and dramatically increased food prices by diverting forests and crops into biofuel production. As intermittent resources, solar and wind raise other issues

Solar energy can be harnessed in different levels around the world. Depending on a geographical location the closer to the equator the more "potential" solar energy is available. Solar plate is a light sensitized steel backed polymer material used by artists as an alternative to hazardous printing techniques. It is a simple, safer, and faster approach than traditional etching and relief printing. It does not use grounds, acids or solvents. it is exposed with u.v. light (the sun) and developed with ordinary tap water.

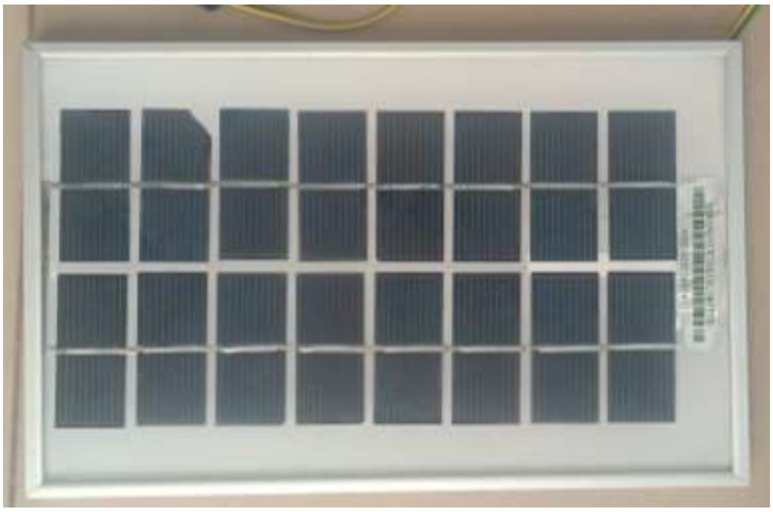


Fig: 2.24 Solar cell/ plate

The opaque materials in the form of non-water based pigments, or... it may be utilized by exposing the plate through a transparent film with artwork on it. The film may be created by drawing on acetate, photocopying or scanning and printing on film, or darkroom techniques with modality a positive transparency is for printing as an etching; a negative transparency is for printing a relief impression. Photovoltaic (PV) cells utilize semiconductor technology to convert solar radiation directly into an electric current which can be used immediately or stored for future use. PV cells are often grouped in the form of “modules” to produce arrays which have the capability to produce power for orbiting satellites and other spacecraft. Recently, with the continual decline of manufacturing costs (declining 3% to 5% per year in recent years), uses of PV technology have grown to include home power generation, and grid-connected electricity generation. Installations of PV systems have also been increasing due in large part to comprehensive incentive programs which help reduce the costs of these systems and also allow users to sell excess electricity back to the public grid (feed-in).

Solar panels generate electricity using the same principles of electronics as chemical batteries or standard electrical outlets. With solar panels, it's all about the free flow of electrons through a circuit.

Evacuated heat pipe tubes are designed such that convection and heat loss are eliminated, whereas Flat-plate solar panels contain an air gap between absorber and cover plate which allows heat loss to occur. Further, thermal heat pipe systems are capable of limiting the maximum working temperature, whereas Flat-plate systems have no internal method of limiting heat buildup which can cause system failure.

Finally, evacuated heat pipe systems are lightweight, easy to install and require minimal maintenance. Flat-plate systems, on the other hand, are difficult to install and maintain, and must be completely replaced should one part of the system stop working.

To understand how solar panels generate electrical power, it might help to take a quick trip back to high school chemistry class. The basic element of solar panels is the same element that helped create the computer revolution -- pure silicon. When silicon is stripped of all impurities, it makes a ideal neutral platform for the transmission of electrons. Silicon also has some atomic-level properties which make it even more attractive for the creation of solar panels.

- This cement helps to prevent the **panel** from becoming too hot, though **solar energy panels** can still overheat. Silicon **solar energy panels** are constructed differently than crystalline **solar energy panels**.
- While years of free **energy** and decreased dependence on fossil fuels may sound great, no examination of **solar energy** pros and cons is complete without a look at

the price, which is considered the largest barrier to conversion to **solar energy** use. **Solar energy panels** and cells are very costly, and several may be required to power a home, business, or vehicle.

2.7.2 Advantages:

- Solar energy is free of pollution.
- The plant requires little maintenance or help after setup.
- It is economical.
- When it is connected to the grid, solar energy can overtake the highest cost electricity at peak demand and can also reduce grid loading, apart from getting rid of the need for local battery power in darkness.

2.7.3 Disadvantages:

- It is available only by day and not when the sky is cloudy, thereby reducing the chances of it being totally reliable and requiring storage facilities.
- It needs a backup power plant to be kept hot and to replace solar power stations as they stop producing energy.
- Keeping backup plants hot includes an energy cost which includes coal burning.
- Places located at high altitudes or those that are often cloudy are not targets for solar power use.
- It can only be used to power transport vehicles by converting energy into another form of energy and recurring an energy penalty.
- Solar cell technologies produce DC power which needs to be converted to AC power, incurring an energy penalty.

Solar energy can be used to generate electricity using photovoltaic solar cells and concentrated solar power, apart from other means. You can use solar power in the house for domestic use.

2.7.4 Applications of solar technology:

Average insulation showing land area (small black dots) required to replace the world primary energy supply with solar electricity. 18 TW is 568 Exajoule (EJ) per year. Insulation for most people is from 150 to 300 W/m² or 3.5 to 7.0 kWh/m²/day.

Solar energy refers primarily to the use of solar radiation for practical ends. However, all renewable energies, other than geothermal and tidal, derive their energy from the sun. Solar technologies are broadly characterized as either passive or active depending on the way they capture, convert and distribute sunlight. Active solar techniques use photovoltaic panels, pumps, and fans to convert sunlight into useful outputs. Passive solar techniques include selecting materials with favourable thermal properties, designing spaces that naturally circulate air, and referencing the position of a building to the Sun. Active solar technologies increase the supply of energy and are considered supply side technologies, while passive solar technologies reduce the need for alternate resources and are generally considered demand side technologies Solar cells can also be applied to other electronics devices to make it self-power sustainable in the sun. There are solar cell phone chargers, solar bike light and solar camping lanterns that people can adopt for daily use. Solar power plants can face high installation costs, although this has been decreasing due to the learning curve. Developing countries have started to build solar power plants, replacing other sources of energy generation.

2.8 LCD display:

LCD Background:

One of the most common devices attached to a micro controller is an LCD display. Some of the most common LCD's connected to the many microcontrollers are 16x2 and 20x2 displays. This means 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively.

Basic 16x 2 Characters LCD

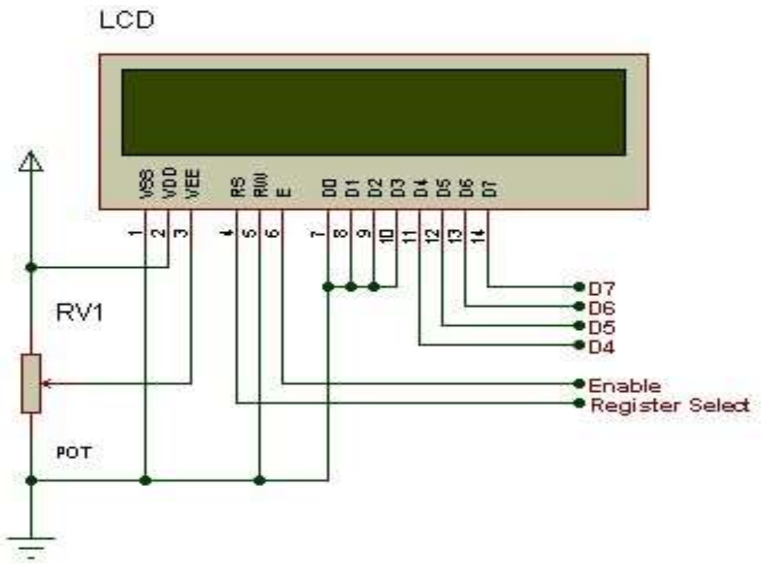


Fig: 2.25 LCD Pin diagram

Table 2.2: Character LCD pins with Microcontroller

Pin description:

Pin No.	**Name**	**Description**
Pin no. 1	**VSS**	Power supply (GND)
Pin no. 2	**VCC**	Power supply (+5V)
Pin no. 3	**VEE**	Contrast adjust
Pin no. 4	**RS**	0 = Instruction input
		1 = Data input
Pin no. 5	**R/W**	0 = Write to LCD module
	EN	1 = Read from LCD module
Pin no. 6	**D0**	Enable signal
Pin no. 7	**D1**	Data bus line 0 (LSB)
Pin no. 8	**D2**	Data bus line 1
Pin no. 9	**D3**	Data bus line 2
Pin no. 10	**D4**	Data bus line 3
Pin no. 11	**D5**	Data bus line 4
Pin no. 12	**D6**	Data bus line 5
Pin no. 13	**D7**	Data bus line 6
Pin no. 14		Data bus line 7 (MSB)

The LCD requires 3 control lines as well as either 4 or 8 I/O lines for the data bus. The user may select whether the LCD is to operate with a 4-bit data bus or an 8-bit data bus. If a 4-bit data bus is used the LCD will require a total of 7 data lines (3 control lines plus the 4 lines for the data bus). If an 8-bit data bus is used the LCD will require a total of 11 data lines (3 control lines plus the 8 lines for the data bus). The three control lines are referred to as **EN**, **RS**, and **RW**.

The **EN** line is called "Enable." This control line is used to tell the LCD that we are sending it data. To send data to the LCD, our program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus. When the other lines are completely ready, bring **EN** high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

The **RS** line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which should be displayed on the screen. For example, to display the letter "T" on the screen we would set RS high.

The **RW** line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD. Only one instruction ("Get LCD status") is a read command. All others are write commands--so RW will almost always be low. Finally, the data bus consists of 4 or 8 lines (depending on the mode of operation selected by the user). In the case of an 8-bit data bus, the lines are referred to as DB0, DB1, DB2, DB3, DB4, DB5, DB6, and DB7.

Schematic:

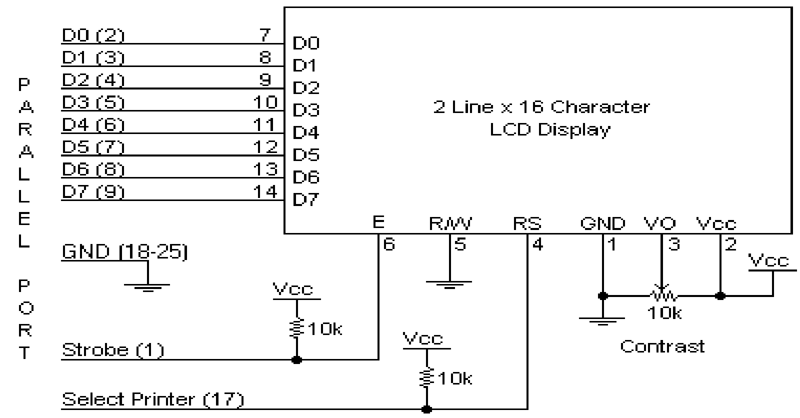


Fig: 2.26 Schematic Diagram LCD display

Circuit Description:

Above is the quite simple schematic. The LCD panel's Enable and Register Select is connected to the Control Port. The Control Port is an open collector / open drain output. While most Parallel Ports have internal pull-up resistors, there is a few which don't. Therefore by incorporating the two 10K external pull up resistors, the circuit is more portable for a wider range of computers, some of which may have no internal pull up resistors.

We make no effort to place the Data bus into reverse direction. Therefore we hard wire the R/W line of the LCD panel, into write mode. This will cause no bus conflicts on the data lines. As a result we cannot read back the LCD's internal Busy Flag which tells us if the LCD has accepted and finished processing the last instruction. This problem is overcome by inserting known delays into our program.

The 10k Potentiometer controls the contrast of the LCD panel. Nothing fancy here. As with all the examples, I've left the power supply out. We can use a bench power supply set to 5v or use an onboard +5 regulator. Remember a few de-coupling capacitors, especially if we have trouble with the circuit working properly.

SETB RW

Handling the EN control line:

As we mentioned above, the EN line is used to tell the LCD that we are ready for it to execute an instruction that we've prepared on the data bus and on the other control lines. Note that the EN line must be raised/ lowered before/after each instruction sent to the LCD regardless of whether that instruction is read or write text or instruction.

In short, we must always manipulate EN when communicating with the LCD. EN is the LCD's way of knowing that we are talking to it. If we don't raise/lower EN, the LCD doesn't know we're talking to it on the other lines.

Thus, before we interact in any way with the LCD we will always bring the EN line low with the following instruction:

CLR EN

And once we've finished setting up our instruction with the other control lines and data bus lines, we'll always bring this line high:

SETB EN

The line must be left high for the amount of time required by the LCD as specified in its datasheet. This is normally on the order of about 250 nanoseconds, but checks the datasheet. In the case of a typical microcontroller running at 12 MHz, an instruction requires 1.08 microseconds to execute so the EN line can be brought low the very next instruction. However, faster microcontrollers (such as the DS89C420 which executes an instruction in 90 nanoseconds given an 11.0592 MHz crystal) will require a number of NOPs to create a delay while EN is held high. The number of NOPs that must be inserted depends on the microcontroller we are using and the crystal we have selected. The instruction is executed by the LCD at the moment the EN line is brought low with a final CLR EN instruction.

Checking the busy status of the LCD:

As previously mentioned, it takes a certain amount of time for each instruction to be executed by the LCD. The delay varies depending on the frequency of the crystal attached to the oscillator input of the LCD as well as the instruction which is being executed.

While it is possible to write code that waits for a specific amount of time to allow the LCD to execute instructions, this method of "waiting" is not very flexible. If the crystal frequency is changed, the software will need to be modified. A more robust method of programming is to use the "Get LCD Status" command to determine whether the LCD is still busy executing the last instruction received.

The "Get LCD Status" command will return to us two tidbits of information; the information that is useful to us right now is found in DB7. In summary, when we issue the "Get LCD Status" command the LCD will immediately raise DB7 if it's still busy executing a command or lower DB7 to indicate that the LCD is no longer occupied.

CHAPTER 3
SOFTWARE DESCRIPTION

This project is implemented using following software's:

- Express PCB – for designing circuit
- PIC C compiler - for compilation part

3.1 Express PCB:

Breadboards are great for prototyping equipment as it allows great flexibility to modify a design when needed; however the final product of a project, ideally should have a neat PCB, few cables, and survive a shake test. Not only is a proper PCB neater but it is also more durable as there are no cables which can yank loose.

Express PCB is a software tool to design PCBs specifically for manufacture by the company Express PCB (no other PCB maker accepts Express PCB files). It is very easy to use, but it does have several limitations.

- It can be likened to more of a toy then a professional CAD program.
- It has a poor part library (which we can work around)
- It cannot import or export files in different formats
- It cannot be used to make prepare boards for DIY production

Express PCB has been used to design many PCBs (some layered and with surface mount parts. Print out PCB patterns and use the toner transfer method with an Etch Resistant Pen to make boards. However, Express PCB does not have a nice print layout. Here is the procedure to design in Express PCB and clean up the patterns so they print nicely.

3.1.1 Preparing Express PCB for First Use:

Express PCB comes with a less then exciting list of parts. So before any project is started head over to Audio logical and grab the additional parts by morsel, ppl, and tangent, and extract them into your Express PCB directory. At this point start the program and get ready to setup the workspace to suit your style. Click View -> Options.

In this menu, setup the units for “mm” or “in” depending on how you think, and click “see through the top copper layer” at the bottom. The standard color scheme of red and green is generally used but it is not as pleasing as red and blue.

3.1.2 The Interface:

When a project is first started you will be greeted with a yellow outline. This yellow outline is the dimension of the PCB. Typically after positioning of parts and traces, move them to their final position and then crop the PCB to the correct size. However, in designing a board with a certain size constraint, crop the PCB to the correct size before starting.

Fig 3.1: Tool bar necessary for the interface

- The select tool: It is fairly obvious what this does. It allows you to move and manipulate parts. When this tool is selected the top toolbar will show buttons to move traces to the top / bottom copper layer, and rotate buttons.
- The zoom to selection tool: does just that.
- The place pad: button allows you to place small soldier pads which are useful for board connections or if a part is not in the part library but the part dimensions are available. When this tool is selected the top toolbar will give you a large selection of round holes, square holes and surface mount pads.
- The place component: tool allows you to select a component from the top toolbar and then by clicking in the workspace places that component in the orientation chosen using the buttons next to the component list. The components can always be rotated afterwards with the select tool if the orientation is wrong.
- The place trace: tool allows you to place a solid trace on the board of varying thicknesses. The top toolbar allows you to select the top or bottom layer to place the trace on.
- The Insert Corner in trace: button does exactly what it says. When this tool is selected, clicking on a trace will insert a corner which can be moved to route around components and other traces.
- The remove a trace button is not very important since the delete key will achieve the same result.

3.1.3 Design Considerations:

Before starting a project there are several ways to design a PCB and one must be chosen to suit the project's needs. Single sided, or double sided?

When making a PCB you have the option of making a single sided board, or a double sided board. Single sided boards are cheaper to produce and easier to etch, but much harder to design for large projects. If a lot of parts are being used in a small space it may be difficult to make a single sided board without jumpering over traces with a cable.

While there's technically nothing wrong with this, it should be avoided if the signal travelling over the traces is sensitive (e.g. audio signals).

A double sided board is more expensive to produce professionally, more difficult to etch on a DIY board, but makes the layout of components a lot smaller and easier. It should be noted that if a trace is running on the top layer, check with the components to make sure you can get to its pins with a soldering iron. Large capacitors, relays, and similar parts which don't have axial leads can NOT have traces on top unless boards are plated professionally. Ground-plane or other special purposes for one side

When using a double sided board you must consider which traces should be on what side of the board. Generally, put power traces on the top of the board, jumping only to the bottom if a part cannot be soldiered onto the top plane (like a relay), and vice- versa. Some projects like power supplies or amps can benefit from having a solid plane to use for ground. In power supplies this can reduce noise, and in amps it minimizes the distance between parts and their ground connections, and keeps the ground signal as simple as possible. However, care must be taken with stubborn chips such as the TPA6120 amplifier from TI. The TPA6120 datasheet specifies not to run a ground plane under the pins or signal traces of this chip as the capacitance generated could effect performance negatively.

3.2 PIC Compiler:

PIC compiler is software used where the machine language code is written and compiled. After compilation, the machine source code is converted into hex code which is to be dumped into the microcontroller for further processing. PIC compiler also supports C language code.

It's important that you know C language for microcontroller which is commonly known as Embedded C. As we are going to use PIC Compiler, hence we also call it PIC C. The PCB, PCM, and PCH are separate compilers. PCB is for 12-bit opcodes, PCM is for 14bitopcodes, and PCH is for 16-bit opcode PIC microcontrollers. Due to many similarities, all three compilers are covered in this reference manual. Features and limitations that apply to only specific microcontrollers are indicated within. These compilers are specifically designed to meet the unique needs of the PIC microcontroller. This allows developers to quickly design applications software in a more readable, highlevel language. When compared to a more traditional C compiler, PCB, PCM, and PCH have some limitations. As an example of the limitations, function recursion is not allowed.

This is due to the fact that the PIC has no stack to push variables onto, and also because of the way the compilers optimize the code. The compilers can efficiently implement normal C constructs, input/output operations, and bit twiddling operations. All normal C data types are supported along with pointers to constant arrays, fixed point decimal, and arrays of bits.

PIC C is not much different from a normal C program. If you know assembly, writing a C program is not a crisis. In PIC, we will have a main function, in which all your application specific work will be defined. In case of embedded C, you do not have any operating system running in there. So you have to make sure that your program or main file should never exit. This can be done with the help of simple while (1) or for (;;) loop as they are going to run infinitely.

CHAPTER 4
PROJECT DESCRIPTION

In this chapter, schematic diagram and interfacing of PIC Microcontroller with each module is considered**.** As shown in fig: 4.1:

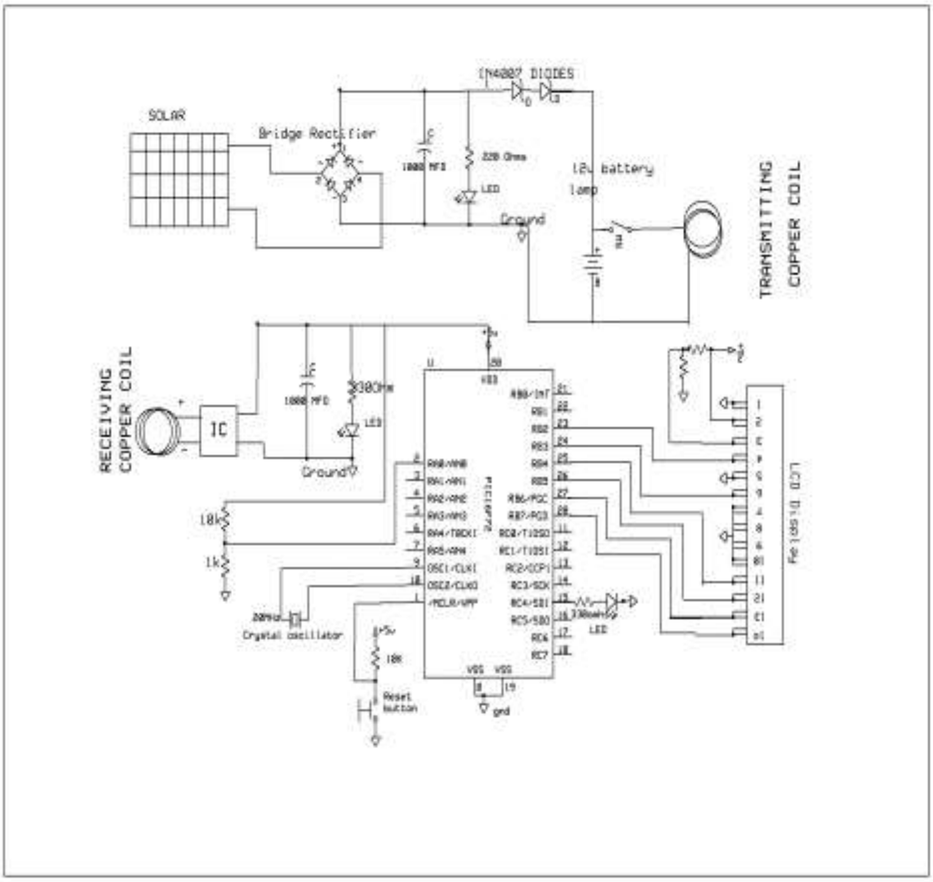


Fig 4.1 schematic diagram of Solar Wireless Electric Vehicle Charging System

The **Solar Wireless Electric Vehicle Charging System** is an innovative solution aimed at promoting sustainable and convenient electric vehicle (EV) charging. This project integrates solar energy harvesting with wireless power transfer technology to create an eco-friendly and user-friendly charging infrastructure.

The system consists of three core components: a **solar power generation unit**, a **power management and storage unit**, and a **wireless charging transmitter and receiver module**. Photovoltaic panels harness solar energy, which is stored in batteries and regulated through an intelligent power management system. The stored energy is then transmitted wirelessly using inductive coupling to the EV, eliminating the need for physical connectors and reducing wear and tear.

This project enhances the practicality of EV usage by enabling wireless charging at parking spaces, public stations, or even residential areas, while simultaneously reducing dependence on grid electricity. It supports the transition to renewable energy and contributes to reducing greenhouse gas emissions.

Key Features:

- Utilization of clean and renewable solar energy
- Wireless charging through inductive power transfer
- Battery management system for energy efficiency
- Environmentally friendly and low-maintenance design
- Potential integration with smart grid and IoT technologies

CHAPTER 5

ADVANTAGES AND DISADVANTAGES

5.1 ADVANTAGES:

- This system helps in transmission of power wirelessly.
- Battery charging circuitry.
- Using solar energy.
- Efficient and low-cost design.
- Low power consumption.
- Easy to operate.
- Easy to install.
- Fast response.
- The advantage of this project is to increase the usage of renewable energy resources in order to reduce the CO emissions.

5.2 DISADVANTAGES:

- Limited distance.

5.3 APPLICATIONS:

- In industries.
- In home appliances.
- Useful for vehicles to charge batteries at tollgates, shopping malls, traffic junctions and roads.

CHAPTER 6

RESULTS

6.1 RESULT:

This project results in a device where the electricity is transmitted wirelessly through copper coils. The system uses solar at the transmitter circuit. Therefore, the current flows from the coil on the transmitter side to the receiver side coil wirelessly while moving the vehicle. This vehicle receives the power from transmitting coil while moving the vehicle. Microcontroller measured voltage from receiving coil will be display on LCD.

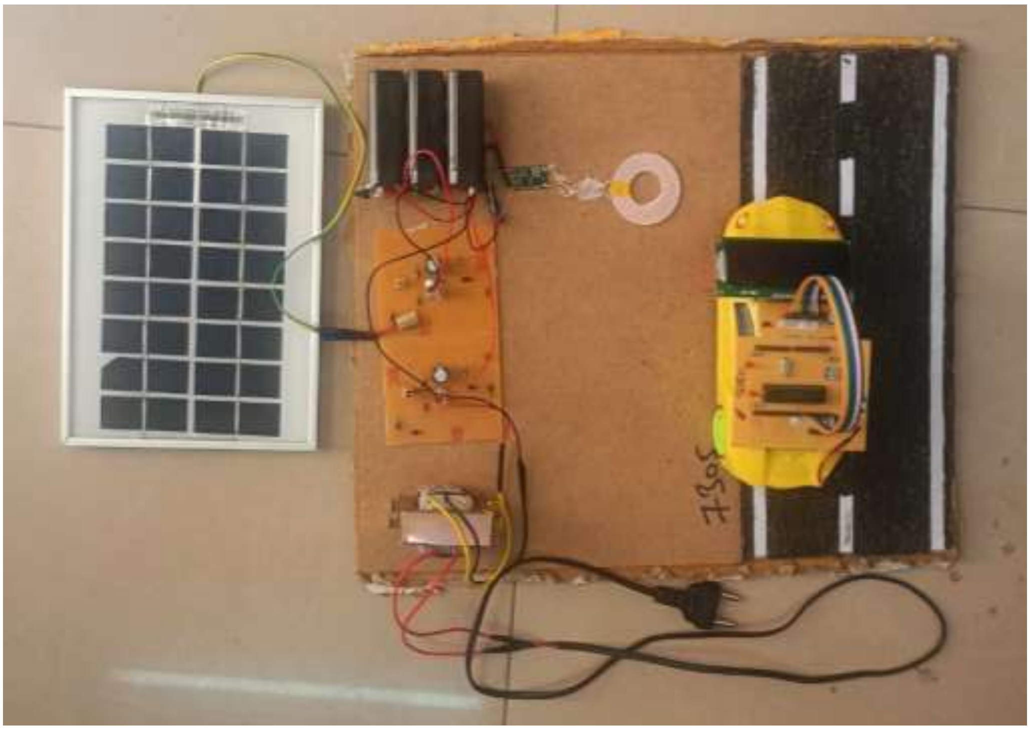


Figure:6.1: Solar Wireless Electric Vehicle Charging System

CHAPTER 7
CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION:

The Solar Wireless Electric Vehicle Charging System represents a significant step toward sustainable and intelligent transportation infrastructure. By combining renewable solar energy with wireless charging technology, the system eliminates the limitations of conventional plug-in chargers and reduces dependence on non-renewable energy sources.

This project demonstrates the feasibility of a clean, contactless, and efficient EV charging method, suitable for modern urban environments and smart city applications. It not only simplifies the charging process but also contributes to environmental conservation by reducing carbon emissions and promoting the use of green energy.

Overall, the successful implementation of this system highlights its potential to revolutionize the way electric vehicles are powered, paving the way for a cleaner and more connected future.

7.2 FUTURE SCOPE:

We can add INVERTER AND GRID setup to this project so, it can work both vehicle to grid and grid to vehicle power converter.

REFERENCES

The sites which were used while doing this project:

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METHODIST COLLEGE OF ENGINEERING & TECHNOLOGY
[Autonomous Institution]
Accredited by NAAC with A+ and NBA
Affiliated to Osmania University & Approved by AICTE

QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)			
Course Name:	PROJECT	**AY**	2024-25
Course Code:	4PW854EE	**Semester**	VIII
Name of the Guide:	Mr. N. NIREEKSHAN	**Class**	EEE
SOLAR WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM			

STUDENTS		
S. No	**HT No**	**Name of the student**
1.	160721734304	G. SREEDHAR
2.	160721734318	M. BHANU PRAKASH
3.	160721734005	KRISHNA CHAITHANYA

Course Outcomes: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	PO
4PW854EE.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	**Apply**	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
4PW854EE.2	Evaluate different solutions based on economic and technical feasibility.	**Evaluate**	PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
4PW854EE.3	Effectively plan a project and confidently perform all aspects of project management	**Evaluate**	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3
4PW854EE.4	Demonstrate effective written and oral communication skills.	**Apply**	PO2, PO5, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3
4PW854EE.5	Develop the hardware	**Create**	PO1, PO2, PO3, PO5, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PS0 1	PSO 2	PSO 3
4PW854EE.1	3	2	2			3		2	2	2	3	3	1	1
4PW854EE.2		3	3		2	2		2		2	2	3	1	1
4PW854EE.3		3	3	3	3	2	2	3	3	3	3	3	3	3
4PW854EE.4		3			3		2	3	3	3	3	1	1	1
4PW854EE.5	3	2	3		2							3	2	3

CO-PO-PSO MAPPING (With Performance Indicators)

PO / CO	PO 1	PO 2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO9	PO 10	PO 11	PS01	PSO 2	PSO 3
4PW854EE.1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2			6.1.1 6.2.1 6.3.1 6.3.2 6.4.1		8.2.1 8.2.2 8.2.3	9.1.1 9.1.2 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	**Slight**	**High**	**High**
4PW854EE.2		2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2		8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1		10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	**Slight**	**High**	**High**
4PW854EE.3		2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2		8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	**Slight**	**Slight**	**Moderate**
4PW854EE.4					5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2			8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	**Slight**	**Slight**	**Slight**
4PW854EE.5	1.1.2 1.2.1 1.3.1	2.1.1 2.1.2 2.2.3 2.4.2	3.1.6 3.2.2 3.4.1		5.1.1 5.1.2 5.2.2							**High**	**High**	**Moderate**

Project Specific Learning Outcomes

LO #	Outcome Description
1	Integrate knowledge from solar energy, electronics, and robotics to design and build a functional solar-powered robot.
2	Analyze and select the most cost-effective components and technically feasible design solutions for the robot.
3	Develop a comprehensive project plan, manage resources, and track progress to ensure timely completion of the robot.
4	Present a technical report and deliver an effective presentation, clearly explaining the robot's design, performance, and results.
5	Design, assemble, and test the hardware components, ensuring the robot performs at least two autonomous functions.

PO Mapping:

Level of Mapping - 3: High; 2: Moderate; 1-Low

PO#	PO Description	Mapped WKs	PO#		Mapped WKs
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Ethics	WK9
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Individual & Collaborative Team Work	
PO3	Design / Development of Solutions	WK5	PO9	Communication	
PO4	Conduct Investigations of complex problems	WK8	PO10	Project Management & Finance	
PO5	Engineering Tool usage	WK2, WK6	PO11	Life-long Learning	WK8
PO6	The Engineer & The World	WK1, WK5, WK7			

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY	7.1.1, 7.2.1	13 CLIMATE ACTION	13.2.2
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING		9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.4.1, 9.5.1	15 LIFE ON LAND	
4 QUALITY EDUCATION	4.7.1	10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION	12.2.1		

QUALITY ANALYSIS

Project collaboration (Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other
Details:							

Project Achievements:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes

S. No	Description	Details
1.	Prototype development	A working prototype of a solar-powered robot was designed, built, and tested using Arduino.
2.	Technology based solution developed	Developed a renewable energy–powered robotic system capable of performing multiple autonomous tasks.
3.	Publication	In Progress
4.	Start-up incubation	----
5.	Competition prizes	----
6.	Community / Industry / HEI service	
7.	SDG Indicators	SDG3, SDG7 (7.1.1, 7.2.1), SDG9 (9.4.1, 9.5.1), SDG11, SDG12 (12.2.1), SDG13 (13.2.2)
8.	Grants / Financial Aid	----
9.	Multi-disciplinary	Involves electrical, electronics, embedded systems, software, mechanical design, and energy systems.
10.	Internship / Employment	----
11.	Self-employment skill	PROSPECTIVE: Technology can be made available to develop a solar-powered automation solutions for rural, smart farming, or surveillance applications.
12.	Use of real-time data	Uses real-time sensor data (e.g., light intensity, obstacle detection) to control robot behaviour and optimize solar energy usage dynamically.

Signature of the Guide **Signature of the Project Coordinator**

Signature of the Assessment Coordinator **Signature of the Head of the Department**